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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION-APRIL 2021

BC406-ANALYTICAL BIOCHEMISTRY-II

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

I. Choose the best answer (10X1=10)

1. Western blotting is a technique to measure _____

   a) DNA b) RNA c) Protein d) Lipid

2. Half life of C-14 is

   a) 10 years b) 3587 years c) 30 days d) 5732 years.

3. Nuclei with atomic number 1 to 82 are

   a) Stable b) Unstable c) Radioactive d) Moderately stable

4. Geiger Muller counter cannot detect

   a) Beta rays b) X rays c) Gamma rays d) Cosmic rays

5. UV-Visible spectrometer uses a prism to...

   a) Focus all wavelengths on the sample simultaneously

   b) Separate radiation into its constituent wavelengths

   c) Reduce the amount of radiation passing through the sample

   d) Stop any radiation going through the sample

   Select the true statement about fluorescence spectroscopy of molecules in

6. Fluorescence probes

   a) Are highly fluorescent reagents that are designed to bind strongly and specifically to certain targets

   b) Are the basis of many very sensitive assays developed biological

molecules of interest

c) Enable the use of fluorescence for the detection of molecules that are not fluorescent

d)All of the above

7. Scintillation counter can detect
   a)beta rays b) X-rays c) gamma rays d) All of the above
8. Absorption occurs at...
   a)All wavelengths in the spectrum
   b) A characteristic wavelength dependent on the molecule
   c) The UV region
   d) The visible region
9. The Beer-Lambert Law...
   a) Relates absorbance, concentration, path length and molar absorption coefficient
   b)Tells us the volume of the sample
   c)Relates frequency and wavelength
   d)Allows us to calculate how conjugated the system is
10. The main advantage of fluorescence over UV-Vis spectroscopy is
    a) Its sensitivity
    b) Its compatibility with separation techniques
    c) Its compatibility with most analytes
    d) None of the above

II. Say true or false (5X1=5)

11. Cosmic rays are ionization radiation.
12. Unit of radio activity is Curie.

13. Monochromatic light is defined as the measurement of the intensity of light at all wavelengths.
14. The most sensitive device used in spectrophotometers for the conversion of polychromatic light to monochromatic is class filter.
15. Specific protein in the sample is detected by Western blotting technique.

III. Answer in one sentence (10X1=10)

16. What lamp is the source of light in Spectroflurimeter?
17. What is the main advantage of fluorescence over UV-Vis spectroscopy?
18. How many types are there in blotting techniques?
19. What are the techniques used to transfer protein onto nitrocellulose paper?
20. What is the common use of RIA?
21. What is a chromophore?
22. Which analytical technique is used to estimate sodium?
23. What radioisotope is used for treating cancer tumors and cells?
24. Define radioactivity.
25. What is the purpose of RIA?

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Give a brief note on Southern blotting technique..
27. Discuss the principle and application of GM counter.
28. Comment on Biological hazards of radiation and its safety aspects.
29. Explain flame photometry – Principle, instrumentation and applications in trace elements.
30. Explain the basic concepts and applications of PCR.
31. What is autoradiography?

SECTION-C (3X10=30)

Answer any THREE of the following:

32. Explain the principle, procedure and applications of RIA.
33. Elaborate on the application of radio isotopes in the field of biology.
34. Explain concepts and applications of Northern and Western blotting techniques.
35. Explain the principle and instrumentation of atomic absorption spectrophotometer with one example.
36. Explain principle, instrumentation and application of spectrophotometer.

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